Enhancing Cybersecurity Awareness and Practices in Rural India: A Framework for Safer Digital Engagement

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***Abstract*— the suggested research paper will explore the awareness and practices of the rural population in Assam, India, with regard to cybersecurity and the changing level of awareness as the digital technologies which are being propagated by the governments via the Digital India program and similar are being spread. Quantitative survey of 231 rural dwellers shows significant disparities in cybersecurity awareness, especially among women and those with lower educational qualification, with over 40% of the participants not able to recognize a phishing attack and 35% of the participants reporting poor password usage habits. Among the obstacles identified in the study, there is a lack of access to digital infrastructure, the lack of community interaction, and the lack of confidence in reporting cyber incidents. The research underlines the need to have culturally sensitive and individualized cybersecurity education; although there are some good results on government-led programs. The following community-based model, based on the idea of putting more emphasis on the local empowerment and education, aims at closing these gaps and improving digital safety, therefore, increasing the resilience to cyber-attack in under proven rural regions.**

Keywords*— *Rural cybersecurity, Digital literacy, Cyber awareness, Digital inclusion, Online safety, Cybercrime risks, Community training

I. Introduction

Background of the Study

India has seen a marked change in the access to the provision of the necessary services such as banking, education, healthcare, and government facilities due to the rapid introduction of digital technology to the country, triggered by such initiatives as Digital India. This growth has brought significant advantages to rural communities and they are able to access essential resources on the internet. However, an increased overdependence on digital platforms has increased vulnerability especially in regions where there is a lack of digital literacy, infrastructure, and awareness programs. Those conditions make rural residents vulnerable to numerous cybercrimes which include phishing, identity fraud and financial fraud. The unequal distribution of cybersecurity infrastructures between urban and rural environments adds to the issue as well because urban areas are more extensive in their resources and support, which contributes to the vulnerability of rural populations to cybercrime (Bissadu et al., 2024; Banda et al., 2007).

The existing cybersecurity in India falls more on an urban-based approach and rural areas are not well represented in national strategies. There is an urgent necessity to determine how rural consumers in Assam are aware of cybersecurity and how they undertake it, thus informing the creation of a relevant cybersecurity framework. The aim of the study is to bridge the knowledge gaps about cybersecurity among people in rural Assam and to develop an effective culturally-appropriate framework to protect online identities and ensure a safe and secure online interaction (Alsmadi and Juma, 2023; Agarwal, 2012).

Importance of the Study

The paper is essential in bridging the cybersecurity gap between rural and urban India. It will provide insights that will be used in creating more effective cybersecurity policies that suit the rural population, thus leaving them out of the digital economy. The study will provide a tailored cybersecurity model that considers cultural, linguistic, and socio-economic factors that will make them more relevant and feasible to communities in rural areas. The research will make the rural people stronger so that they can protect their online identities and utilize the digital services without fear of being victims of cybercrime and thus, minimize their vulnerability by creating awareness of digital safety (Digital Empowerment Foundation, 2024 & Fuchs, 2023). Moreover, the results can make the necessary services, including online banking, health care, and education, more accessible, as well as increase the economic and social life of rural areas. The current study does not only add to the current body of knowledge on cybersecurity in rural areas, but also presents a practical example of how similar interventions may be applied in other developing areas of the world. The research highlights how local engagement can be used in promoting long-term sustainable digital safety practices through the use of community based solutions.

Scope of the Study

The rural Assam region of India is the location of the current study as the region has reached significant levels of digital inclusion due to the governmental efforts including Digital India

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(Gogoi et al., 2024). However, even with this increased connectivity in the digital realm, the rural population still faces a set of challenges that are unique to them, which hinder the safe usage of digital platforms. The area of this study will include several aspects of the critical aspects of cybersecurity in rural Assam:

- Cybersecurity Awareness: The research will measure the level of awareness of the prevalent cyber threat, which are phishing, identity theft, and financial fraud. It will look at the level of familiarity of rural residents with generic security practices, such as password management, identifying suspicious links, and knowledge of online scams.
- Digital Behaviors and Practices: The research will examine the daily online behavior among the rural population, using both mobile phones and internet-based services to carry out banking, education and even healthcare. It will examine their approach to cybersecurity, or lack thereof, and points of vulnerability in their online security approaches (Digital Empowerment Foundation, 2024).
- Barriers to Safe Digital Engagement: The study will be used to determine the key obstacles to the implementation of safe digital practices, such as low digital literacy, lack of access to secure digital technologies, ineffective internet connectivity, as well as the lack of cybersecurity education specific to rural communities. The analysis will also focus on cultural variables including gender and education differences that determine cybersecurity behaviors.
- Demographic Analysis: The study shall be used to interpret the impact of demographic factors such as age, sex, and education level, and occupation on cybersecurity behavior by rural dwellers. It will specifically focus on gender-related barriers because in rural areas women are often faced with more challenges in trying to seek digital literacy programs and cybersecurity awareness content.
- Development of a Cybersecurity Framework: As per the results, the research will seek to develop a feasible and cultural-sensitive cybersecurity model in rural Assam. The framework will also suggest the tailored educational resources, community-based interventions and affordable cybersecurity equipment to empower the rural communities to protect their online identities.
- Focus on Assam: Although the problem of digital safety is widespread in the whole of rural India, Assam is chosen as a case study in the present paper. The variety in digital infrastructure and the level of digital literacy in Assam makes it a perfect location to study the needs and the difficulties of rural residents in relation to digital safety.

II. Literature Review

Rural India has been changing significantly in the way the communities there access essential services, as a result of the digitalization of the rural areas through various projects such as Digital India. With the rise in internet penetration, there is an improved access to banking facilities, education, and healthcare and government services by rural population. Nonetheless, rural dwelling people are also facing cybersecurity risks due to this increased dependency on digital platforms. Rural populations are extremely vulnerable to the cybercrimes of phishing, identity theft, and financial fraud due to the lack of digital literacy, insufficient infrastructure, and poor understanding of the dangers of cybersecurity risks. Besides, these vulnerabilities are worsened by the lack of cybersecurity support in the rural communities, as compared to the urban centers. The cybersecurity gap between urban and rural areas is an urgent issue because residents of the rural areas still have to deal with significant threats to their digital identities (Ahsan, 2024; Alsmadi, Abassi, and Singh, 2023).

A. Digital Literacy and Cybersecurity Awareness

Digital literacy is the core of cybersecurity awareness and is the first step in protecting the rural population against cyber-threats. One of the ways in which cybersecurity awareness can be improved significantly is through addressing digital literacy gaps. As an example, culturally relevant and context-dependent bespoke digital education programs are needed to enable rural residents with the ability to manoeuvre within digital spaces without compromising. The various levels of literacy that are common in rural areas should be reflected in these programs, considering linguistic and educational background. Effective local campaigns have proven that the specific programs can foster awareness significantly and reduce the risk which is inherent in using the Internet, such as poor quality passwords and poor understanding of the threats of the Internet. However, this has not been effective in dealing with more advanced threats like phishing and identity theft which are less understood. Based on that, the training should not be confined to theoretical training, but also includes practical and real-life situations to provide rural residents with the ability to identify and respond to these complex threats in a better way (Ahsan, 2024; AISECT, 2024; Chatterjee et al., 2022).

B. Impact of Social and Gender Dynamics on Cybersecurity Practices

The social aspect of interaction with gender roles has a strong impact on cybersecurity in rural India. Access to cyber security and other barriers to digital literacy especially among rural women are often faced by them in addition (Heeks and Ospina, 2019). It has been empirically observed that females in rural societies are less digitally literate and tend to take a more passive approach to cybersecurity measures. This is because the social norms hold onto this gender inequality as women have limited access to technology and formal education. Women, therefore, have a lower propensity to be aware of the dangers of cyber threats like phishing and online frauds (Alsmadi, Abassi, and Singh, 2023; Basri, 2018). To correct these inequalities, there is a need to adopt gender sensitive education programs and community based programs. Besides, men and women in rural regions need specific training programs that would address their unique needs and challenges. These campaigns are to produce trust towards the digital tools and emphasize the need to adopt safe digital behavior. With the inclusion of women into such community-based cybersecurity programs, the general awareness and adoption of safe online behavior can be increased in rural communities (Bhardwaj, 2023; Ambika and Nair, 2023).

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C. Technology Adoption and Its Impact on Digital Security

The use of online platforms, such as mobile wallets and electronic health services has minimized hurdles to needed resources in the rural region. However, with this increased dependence on digital assets, cybersecurity-related issues have also emerged in a major way. Since the uptake of technology among rural residents is at a high rate, there are often problems with maintaining the security of online activities (Basri, 2018; Banda et al., 2007). These populations are very susceptible to cyber threats due to the lack of digital literacy and the lack of cybersecurity knowledge. Specifically, the unclear state of regulations and cyber laws, as well as the absence of their enforcement, aggravates the problem of personal information protection. Mobile banking, e-commerce and e-health services have also led to the vulnerability of rural dwellers to cybercrime and many rural dwellers are still ill equipped to handle such threats. As can be observed, in as much as the adoption of technology has given the rural communities the upper hand in several ways, it has also brought with it new vulnerabilities which need to be mitigated by various means through integrated digital literacy and cybersecurity information and training programs (Agarwal, 2012).

D. Data Sharing Practices and Privacy Concerns

The use of smartphones in India rural areas has significantly revolutionized the data sharing culture with its opportunities and threats. Facebook and WhatsApp are social media platforms that are widely used to publish personal information, videos, and photographs. Although these platforms are beneficial as they allow communicating and building communities and are simultaneously exposing users to significant privacy risks, particularly when they are not aware of how to manage their privacy settings (Basri, 2018; Bissadu et al., 2024). The comfort of sharing information on these sites has intensified the issues regarding the privacy of the data since the rural inhabitants often cannot afford to protect the personal information that is sensitive. Data privacy education is a necessary factor to reduce the risks of excessive sharing of personal data. Also, lack of education in regard to privacy settings makes the rural communities vulnerable to identity theft, online fraud and unfair use of data. As such, there should be educational programs aimed at educating rural users on how to use the privacy settings, how to identify malicious links, and how to protect their personal information to reduce such risks (Bhardwaj, 2023).

E. Cybersecurity Frameworks and Community-Based Approaches

Development of cybersecurity systems to be used in rural India requires the fact that such communities have unique socio-economic and cultural backgrounds. Global cybersecurity norms often do not concern rural communities, which overlook such peculiarities of the population as low digital literacy, inadequate infrastructure, and cultural differences (Chatterjee et al., 2022). To make these structures more representative of the rural setting, there is a need to involve local leaders and community influencers, who also can become cybersecurity ambassadors. Locally defined methods of cybersecurity have proven to be effective in spreading information and creating secure online activities. The success of these programs is however based on long term training and capacity building. Rural populations need long-term resources and maintenance to improve future cybersecurity education opportunities via local leaders, teachers, and influencers. The programs promote practices related to cybersecurity and contribute to increased awareness of cybersecurity, thus creating trust in the communities and providing culturally relevant content (Alsmadi and Juma, 2023; Bissadu et al., 2024).

Many challenges prevent the rural population to become secure users of digital practices. Regulatory loopholes, inability to afford, affordable cybersecurity, and lack of government involvement are among the most influential problems (Manzar and Srivastava, 2022). To overcome these obstacles, the government ought to use a multi-polar approach, which includes offering cheaper cybersecurity services, developing awareness programs, and provision of digital literacy resources in rural areas.

Besides the awareness programs, there should be practical training on issues relating to password protection, phishing and safe sharing of data. Rural inhabitants should also be empowered with the capacity to detect and counter cyber threats to avoid their susceptibility to cyber-attacks. The collaboration between government agencies, NGOs, and local authorities is the way to overcome these obstacles and create the environment, in which the rural population will be able to use digital services in a safe way (Agarwal, 2012; Bissadu et al., 2024).

III. Methodology

The proposed research will be conducted using the quantitative descriptive research design, which is going to investigate cybersecurity awareness, behaviors, as well as concerns of the rural population in Assam, India. The current paper aims to clarify the attitude and mitigation measures embraced by the rural residents regarding cybersecurity risks, especially with the growing access to online services in key areas like banking, health, and education. The increasing rural population that is increasingly becoming digitally exposed is a great opportunity to explore the ways in which these demographics are using and protecting themselves against the new online risks (Chatterjee et al., 2022; Kalan, 2024).

A. Research Design and Approach:

The questionnaire survey becomes the most appropriate mode of data collection in this study. The strategy helps to conduct a quantitative methodological evaluation of many variables, such as digital practices, cybersecurity awareness, perceived risk, and security behavior. The survey was designed based on the well-founded theoretical models, namely, the Technology Acceptance Model (TAM) and the Diffusion of Innovation (DOI) theory. The models also guided the formulation of the questionnaire by providing theoretical frame of reference with which to understand how age, education level, and gender, together with technological variables, affect cybersecurity habits of the rural participants. These frameworks will ensure that the findings will offer a stringent account of the

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influence of different factors on cybersecurity behaviors of the rural communities (Agarwal, 2012).

B. Population and Sampling:

The authors have focused on the group of adult people aged eighteen and above living in the rural and semi-urban Assam and using the internet resources on a regular basis. The respondents were chosen to comprise a sample of 231 participants, thus giving the data statistical reliability and analysis appropriateness. The sample size was carefully identified to level the statistical power against the logistical feasibility to ensure that the sample is large enough to be representative and still manageable considering the limitation of rural environments. The convenience sampling was used because the rural population is scattered and thus it is hard to access all the people. The chosen group was also heterogeneous, including individuals belonging to various age groups, genders, educational levels, and professions and, therefore, making the analyzed rural population relatively comprehensive in terms of cybersecurity behaviors (AISECT, 2024). It was also this heterogeneity that enabled a wide range of perspectives to be captured hence making the findings more generalizable.

C. Recruitment and Data Collection of the participants:

To get the participants, the survey was distributed online, with the help of local institutions and community leaders to make sure that the majority of the population took part in the survey especially where people have low internet access. In some of the areas where internet penetration was not high, the help of some institutions in those areas was invaluable to access potential respondents. The questionnaire was translated to several regional languages to ensure inclusiveness and hence addressing respondents of different literacy levels and rendering the questions to be culturally relevant. In line with the ethical research standards all the participants were assured of the study nature and consent procedures before they participated in the study. Those who found it hard to navigate through the survey were helped to overcome the access barriers, thus giving everyone equal chances of participating in the exercise regardless of their level of technological literacy (Basri, 2018). This method was crucial towards ensuring that the sample was representative and inclusive.

D. Survey Instrument:

The survey tool included 29 questions, grouped into five general areas: demographic data, using and accessing digital devices, knowledge of cybersecurity-related risks (e.g. phishing and financial fraud), cybersecurity behavior (e.g. password strength and reporting a cybersecurity incident), and knowledge of local government policies and the existing cybersecurity training. The respondents were measured in the degree of their agreement with certain statements by the use of a Likert scale, which allowed reaching a more in-depth insight into their attitudes and behaviors as they apply to cybersecurity. This scale made it possible to capture subtle answers that capture the degree of concern, awareness, and proactive behavior of the participants. The questionnaire was designed in a way that was more focused on investigating not only the general awareness but also on those protective measures that are taken by local people in the digital space (Pandey, 2024). By using this methodological tool, the research study attempted to know the gaps as well as the strengths in cybersecurity practices and knowledge among the rural population.

E. Data Analysis:

Statistical software such as SPSS and Excel were used in the data cleaning and coding. General cybersecurity behavior and demographic statistics were summarized using descriptive statistics, frequencies, percentages, means, and standard deviations. The relationships between demographic factors (e.g. the level of education, gender and age) and cybersecurity behaviours were analysed with the help of inferential statistics (Heeks & Ospina, 2019). The correlation test was also applied to examine the relationship between the aspects of digital literacy and safe online behaviors. Comparative analysis was carried out using cross tabulations between various groups of respondents.

IV. Results

This paper has discussed the cybersecurity awareness and behavior of rural villages in Assam, in the sense of finding out major demographic disparities and gaps that need to be addressed. The analysis shows that gender, education, location, and internet proficiency are factors that can impact the behavior of cybersecurity, but age did not show any significant change.

A. Demographic Differences in Cybersecurity Behavior

Based on chi-square analyses, it was revealed that gender, education, and location have a strong effect on cybersecurity practices. It was found that men were more familiar with data risks, phishing, password management, and online scams as compared to women. These results can be used to support the presence of a gender gap in cybersecurity awareness, which, in turn, justifies the need to conduct specific training interventions among the rural women to meet their unique needs.

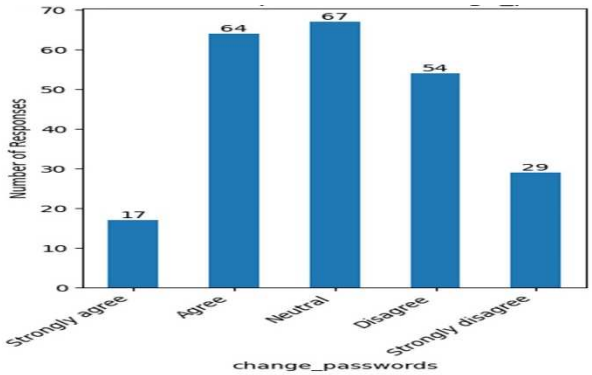


Fig. 1. Distribution of Responses for change password

Fig. 1 demonstrates the way respondents answered the question about changing passwords. The statistics show that the percentage of those who were indifferent to the frequency of changing passwords is a large percentage of 67; this shows that there is a lack of password management activities. Only 64

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participants stated that they change their passwords frequently, and 29 did not, and it is necessary to provide more knowledge about strong passwords.

B. Location and Security Tool Use

The use of security-related applications and the discretion used in sharing personal information portrayed a strong difference depending on the geographical area. City residents were more likely to use security means and take care in sharing personal data in comparison to rural residents. The mismatch highlights the existence of digital divide since people in rural areas have a relatively lower exposure to various cybersecurity protections and tools.

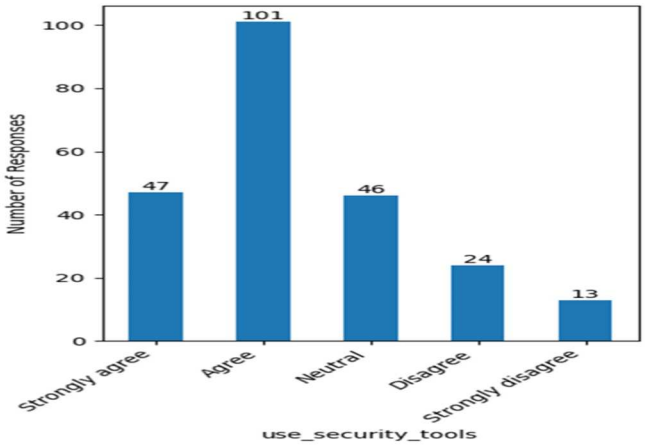


Fig. 2. Distribution of Responses for use security tools

Fig. 2 shows a break-down of the responses on the use of security tools. Based on the results, 101 respondents said that they agreed, and 47 respondents said that they strongly agreed. On the contrary, 24 respondents did not agree, and 13 respondents strongly disagree, which shows inconsistencies in the use of security tools in the rural population. These findings support the need to employ geographical solutions in facilitating the use of simple security measures among rural population.

C. Internet Skill Level and Safe Online Behaviors

Internet proficiency was established to be a major determinant of safe internet behavior in relation to password control, and prevention of online security threats. The ANOVA and the regression analysis proved that more internet skilled participants were more likely to participate in safe internet activities, that is, using strong passwords and avoiding risky internet behavior. This brings out the importance of digital literacy in enhancing cybersecurity awareness.

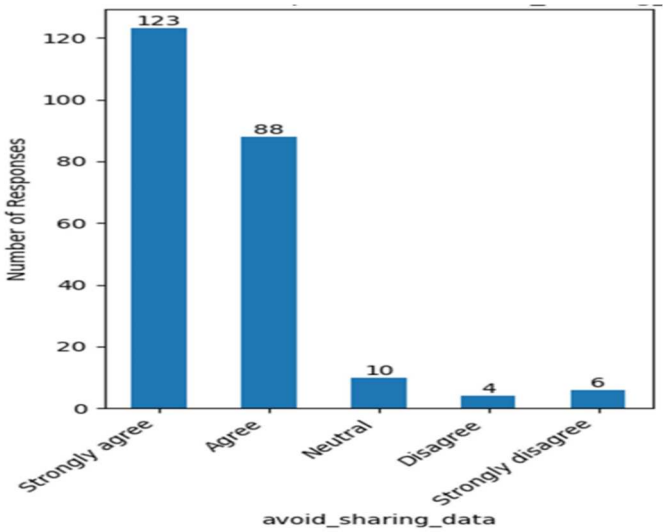


Fig. 3. Distribution of Responses for avoid sharing data

Fig. 3 shows the distribution of the answers in relation to avoiding sharing of personal data. Most of the respondents (123) strongly agreed with not sharing data, and only 88 agreed with the same practice. However, 10 of them were neutral, and few (6) disagreed. These results imply that despite the overall understanding of risky behavior in regard to sharing personal information, there is still a need to work on it, especially in promoting a stable healthy behavior of sharing data.

D. Confidence in Online Security

They trusted the safety of virtual surroundings, which was one more important factor defining cybersecurity behavior. Those people who demonstrated the higher level of the trust in the information about cybersecurity transmitted through the local bodies and governmental agencies were more likely to use secure online behavior. Specifically, the efforts by the government were very crucial in building confidence in online security.

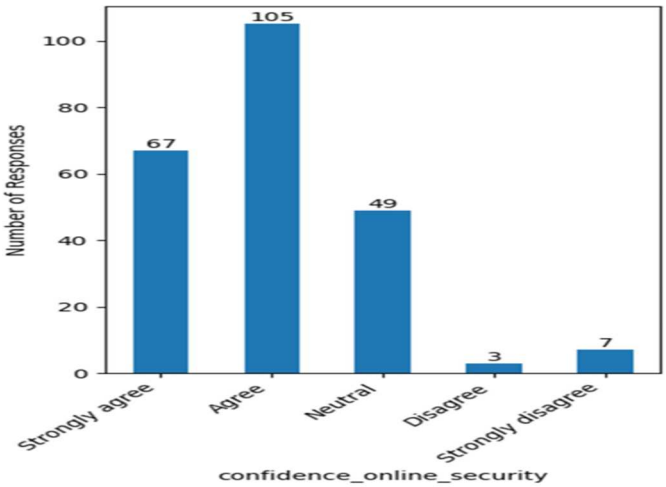


Fig. 4. Distribution of Responses for confidence online security

The distribution of the response about the confidence in online security is shown in Fig. 4. This indicates that a total of 105 respondents agreed that they are confident in their ability to handle online security matters and 67 respondents strongly agreed. Even though most of the respondents respond with

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confidence, there is a relatively minor yet substantial proportion of respondents who strongly disagree, that is, seven respondents, which also suggests that a group of individuals do not trust online security measures. This highlights the need to introduce more trust-building projects in the rural setting, particularly those that will create more credibility to digital safety programs.

E. Government Digital Safety Initiatives

Government-led digital safety initiatives have had a positive impact on the cybersecurity behaviors of rural residents. However, the effectiveness of these programs could be further enhanced by targeting the least engaged individuals, who remain the most vulnerable due to limited cybersecurity knowledge and trust in local information sources.

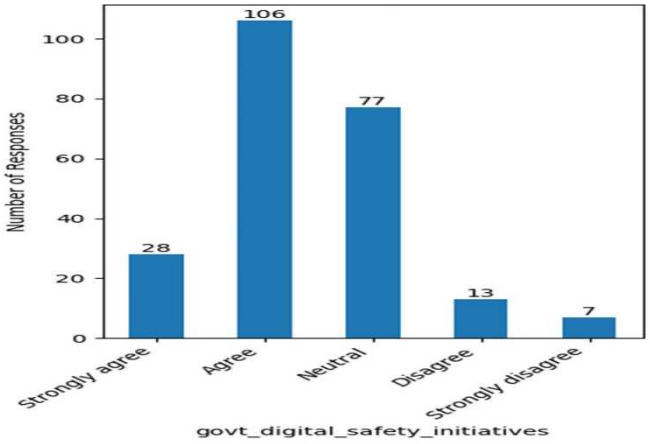


Fig. 5. Distribution of Responses for government digital safety initiatives

Fig. 5 is a visual representation of the distribution of the responses related to the governmental digital safety efforts. Most of the people who responded (106) supported the efficacy of such initiatives whereas 77 were neutral and 28 strongly agreed. However, 13 respondents did not agree and 7 strongly did not agree, which means that though some effect has been produced by the use of government programs, their scope and efficiency should be improved in order to address a wider audience.

F. Engagement Levels and Vulnerability

The sample was categorized into three groups according to the extent of their interactions with cybersecurity, high level of engagement, moderate level of engagement and least level of engagement. The most vulnerable group was the least involved group, which has a poor understanding of cybersecurity and a lack of trust in the local sources of information. They are less likely to take preventive actions to counter the threats they face online and this is where the need to provide specific and easily available interventions that would address this vulnerable population is in place.

G. Focused Interventions and Recommendations

The outcomes of this research point to the need to have specific pragmatic interventions in order to increase cybersecurity awareness in rural Assam. The efforts must prioritize community engagement, promote affordable security tools provided by the government, and promote trust in locally distributed information on cybersecurity. Individualized measures should also be implemented to correct specific weaknesses particularly in women and the least involved. These interventions cannot be ignored in bridging the existing knowledge gaps and behavior among the rural population.

Table 1 Cybersecurity Behavior Outcomes by Predictor

Predict-or	**Outco-me**	**df**	**Chi-square**	**p-value**
Internet access type	Change passwor-ds	21.6471	0.0417	Signifi-cant
	Avoid sharing data	51.2350	8.47e-07	Signifi-cant
Internet access type	Use security tools	20.3254	0.0612	Near-signific-ant
Internet usage purpose	Avoid sharing data	49.4956	0.00026 1	Signifi-cant
Internet frequen--cy	Understa-nd	35.7031	1.99e-05	Signifi-cant

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	online scams			
Internet skill level	Change passwor d-ds	29.7259	0.0195	Signifi-cant
Internet access type	Change passwor d-ds	21.6471	0.0417	Signifi-cant

These statistical findings in table 1 clearly show that there is a strong correlation between various behavioral outcomes, such as changes of passwords, not sharing data, and knowledge of scam, and various variables, including the type of internet access, purpose of use, frequency, and level of skill.

The paper finds there is significant evidence of lack of knowledge and awareness on cybersecurity among the rural population in Assam. Even though most of the respondents are aware of cybersecurity risks, such as phishing, identity theft, and financial fraud, they do not always take preventative measures. Though a few of the respondents are aware of good passwords and security software, they do not use them regularly, mostly due to convenience, and habitual nature. As a result, there exists an apparent demand of full-scale, interactive cybersecurity education that focuses on phishing education, password management, and cybercrime reporting procedures.

Demographic factors such as gender, level of education, and location influence the behavior with regards to cybersecurity. Male individuals, urban dwellers and highly educated individuals demonstrate better cybersecurity practices, where women individuals, rural dwellers and the low educated individuals show relatively low levels of participation in safe online behavior. This focus highlights the need to have comprehensive cybersecurity programs that can be used to assist rural communities and women.

There exists evidence of the success of community-based initiatives, which is more culturally oriented, and is delivered by the local leaders and leaders to enhance cyberspace security confidence and awareness. Nevertheless, the concern about the access to and the relevance of training tools still does exist. Various respondents have expressed their firm belief in continuous training in cybersecurity and life-long learning as a way of keeping pace with changing threats.

Assam, in general, and Rural Assam, in particular, needs an urgent approach to cybersecurity which should include both governmental and community efforts as well as affordable security measures. The deficit can be countered successfully with the integration of local resources with the federal programs and the development of the available reporting mechanisms. The knowledge obtained through this study can be a valuable source of information to policymakers, Internet Service Providers, and leaders in communities to develop effective interventions that will enable the rural population to overcome cyber threats.

V. Conclusion

The paper has explored both the awareness and practices of cybersecurity in the rural Assam region, India, and found gaps in the digital safety behaviors. Despite the awareness of cybersecurity risks, the regular use of protective measures has not spread. The findings show that the government-organized campaigns and dependency on local sources of information are critical to promoting protective behaviors. However, community-based programs should be culturally more relevant and interesting. The use of security tools which are cheap is also poor despite their availability especially to the rural population and women who have less tendency to engage in secure practices. The paper highlights the importance of a more vigorous government intervention, communities involvement, and increased access to easy and less expensive security tools to create less unsafe online space. An integrative approach that involves the government, communities, and available resources is unavoidable to the promotion of cybersecurity behaviors and protection of digital identities in rural environments.

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